Crowded Out: Measuring Rental Market Pressure with Household Density

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Abstract

We introduce the Rental Density Index (RDI), a household-level measure of rental market tightness defined as the number of occupants per renter household. We show that values, changes, and variations in renter-household density contain predictive information about future apartment rent growth across U.S. metropolitan areas. Interpreted within a stock-flow adjustment framework, the level of renter household density functions as a slow-moving state variable that summarizes accumulated imbalances between household formation and housing supply. Using a panel Autoregressive Distributed Lag (ARDL) model, estimated on data for the 100-largest U.S. metropolitan areas, we find that markets with above-average crowding experience stronger subsequent real rent adjustment, even after conditioning on new supply, lagged rent growth, bedroom-density, and fixed effects for time and metropolitan area. The predictive content of the RDI is absent in placebo tests, and it is robust to autocorrelation, reinforcing its interpretation as latent demand pressure rather than a mechanical corollary. Our results highlight household consolidation as an important non-price margin of adjustment in rental markets and demonstrate that renter-household density provides information about market tightness not captured by prices, vacancies, or absorption alone.

Keywords: density, latent demand, multifamily housing, forecasting, rent growth

1 Introduction

Housing availability and affordability has become a central policy concern in many U.S. metropolitan areas. Over several decades, housing production has grown slower than population, contributing to a national housing shortfall. While the median estimate of this shortfall is 3.9 million units, recent estimates range widely–from 1.5 million to over 20 million units–reflecting differences in methodology and underlying assumptions (Table A1). Consistent with these pressures, a substantial share of renter households devote a large fraction of income to housing: recent Census estimates indicate that nearly half of renter households spend more than 30 percent of income on rent, a commonly used threshold for defining housing-cost burden U.S. Census Bureau (2022). Over the same period, shelter costs rose faster than overall consumer prices; between 2000 and 2024, the CPI for shelter increased by 107 percent, compared with a 71 percent increase for non-shelter items Federal Reserve Bank of St. Louis (2024). Despite these national trends, several high-growth metropolitan areas have recently experienced rent declines following periods of rapid new construction Mott (2024). Together, these patterns indicate that rent outcomes reflect both long-run supply constraints and short-run market adjustments that vary across locations.

Measures commonly used to characterize demand in multifamily housing markets, such as occupancy rates and net absorption, are informative about near-term conditions but are closely tied to the existing housing stock. Occupancy rates are mechanically capped, and net absorption primarily reflects the pace at which newly delivered units are leased rather than the intensity of underlying household formation pressure, particularly when supply adjusts slowly Mueller and Laposa (1999). As a result, these indicators provide limited insight into how demand is expressed in markets where households respond along other margins, such as by altering living arrangements Gabriel and Nothaft (2001); Pyhrr, Cooper, and Wofford (1999); Sirmans, Sirmans, and Benjamin (1991). This limitation complicates efforts to distinguish whether observed rent movements primarily reflect supply expansion, changes in household formation behavior, or shifts in the desirability of particular locations Molloy (2022); Pennington (2021). The distinction, however, is central to ongoing debates over the relative roles of supply constraints Saiz (2010) and demand for high-amenity locations Davidoff (2015).

This paper studies rental markets through the lens of a stock-flow adjustment process in which housing supply evolves slowly, while population and household formation respond more quickly to economic conditions. Within this framework, we focus on renter household density (called the Rental Density Index, or, RDI), defined as the average number of occupants per renter household in a metropolitan area. Changes in household density reflect how renters adjust their consumption of housing space in response to changes in prices or availability. When housing becomes scarce or expensive, households may delay forming independent households, they may add roommates, or they may consolidate living arrangements. When conditions ease, households tend to de-densify through expansion. These adjustments can occur even when measured occupancy remains high, making the RDI a potentially informative indicator of space-market pressure.

Variation in renter household density reflects both long-run structural factors and short-run market imbalances. Age composition, family structure, and unit size shape baseline household sizes across metropolitan areas, while departures from these fundamentals arise when supply does not adjust quickly to shifts in demand. We use fixed effects and bedroom-density to account for these metro-level differences, and we recover the impact of changes in RDI, summarizing latent demand pressure within a stock-flow adjustment framework. In this paper, we use the term "latent demand" in a narrow sense: not as unmet preferences at prevailing prices, but as accumulated pressure arising when population and household formation adjust more quickly than housing supply. Above-average renter household density reflects how this pressure is temporarily absorbed through household consolidation rather than immediate price adjustment.

This perspective is motivated by broader changes in population density, rental costs, and tenure choice over recent decades. Between 2000 and 2022, population density in high-income countries increased by approximately 12 percent, while density in the United States increased by roughly 19 percent The World Bank (2025). Over the same period, rents rose faster than both wages and non-shelter prices Federal Reserve Bank of St. Louis (2024); Feiveson, Levinson, and Wertz (2024), and younger cohorts increasingly delayed homeownership and remained in the rental sector longer Fannie Mae (2023). These trends suggest that household formation and space consumption have become important margins through which rental markets adjust.

Prior research in urban economics and real estate has examined density primarily in geographic terms (e.g., persons-per-square mile) and linked it to wages, rents, and productivity through agglomeration effects Liu, Pan, and Li (2018); Titman, Wang, and Yildirmaz (2024). Our focus differs by emphasizing density at the rental household level, which directly reflects how many people share a given housing unit. Because household-level density can change even when geographic density and occupancy remain stable, it provides a complementary perspective on demand pressures in constrained markets.

Building on standard models of housing demand in which households value space but face budget constraints Molloy (2022); Muth (1969), we test whether excess crowding predicts subsequent rent adjustment. Using panel data for the 100 largest U.S. metropolitan areas, we estimate an ARDL specification in which the lagged level of RDI plays a role analogous to an error-correction term. We find that crowding beyond demographic and structural fundamentals is associated with higher subsequent rent growth, consistent with a stock-flow adjustment process in which prices respond to latent demand pressures over time. Figure 1 illustrates the relationship between excess crowding and future real rent growth.

While the majority of our analysis focuses on the level of household density, we also document empirical relationships between the change in Rental Density Index (RDI) and subsequent rent dynamics. Table 1 shows the near-term effects of densification or expansion, conditioned on starting density. The observed patterns in the level and change of RDI are consistent with the Wheaton-DiPasquale four-quadrant model. In high-density markets, periods of de-densification (declines in RDI) reflect the release of pent-up household formation into scarce space: households that had consolidated under pressure are now able to form independent units, absorbing available inventory and driving stronger subsequent rent growth (17 bps) relative to continued consolidation (6 bps). By contrast, in low-density markets, both regimes are associated with near-zero or negative rent outcomes: de-densifying markets average -5 bps,

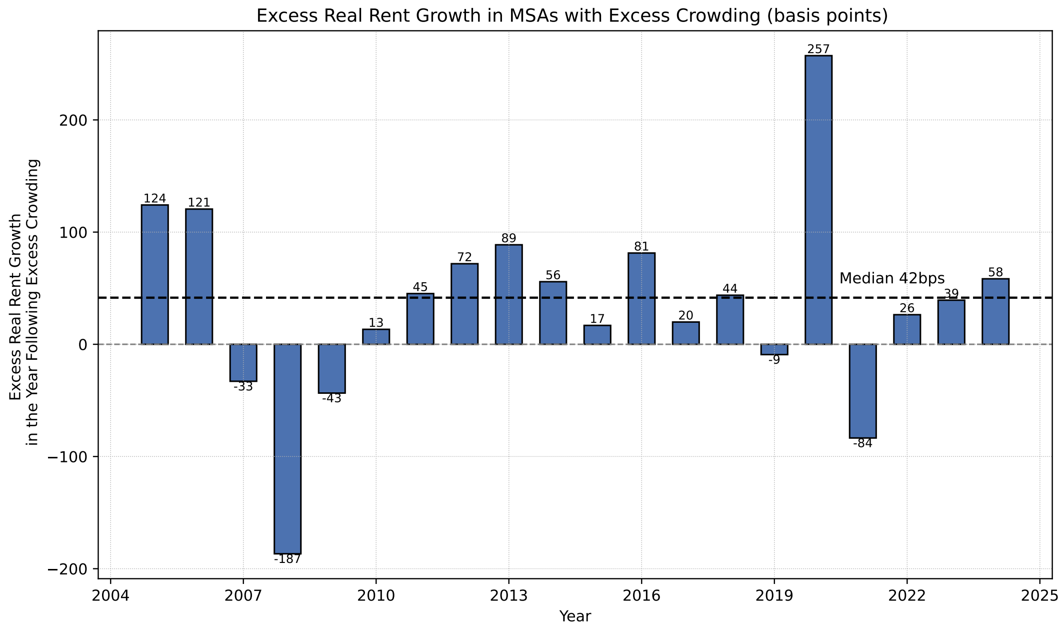

Fig. 1: Difference in means of relative real rent growth: markets with and without excess crowding. Excess crowding is defined as having a renter household density greater than the national average total household density. Excess crowding is evaluated each year. The rent growth is measured the following year. The data is from the 100 largest Core-Based Statistical Areas (CBSA) for each year between 2005 and 2025.

while densifying markets average approximately zero, consistent with slack absorbing adjustment pressure before it reaches prices. Although changes in RDI can arise mechanically through changes in either renter population or renter households, declines in rental unit counts are rare in our sample, occurring in fewer than 50 of 2,000 CBSA-year observations. And we explicitly control for supply changes in our empirical specifications.

This paper makes three principal contributions. First, we introduce the Rental Density Index (RDI), a household-level measure of rental market tightness defined as the number of occupants per renter household. Unlike price- or vacancy-based indicators, RDI captures the non-price margin of adjustment through which rental markets absorb demand pressure via household consolidation, providing a signal of latent demand that is not reflected in observed rents or absorption rates.

Second, we embed RDI in a stock-flow adjustment framework and show empirically that above-average crowding functions analogously to an error-correction term: markets where household density exceeds the contemporaneous national average experience stronger subsequent real rent growth, even after conditioning on new supply, bedroom-density, and lagged rent dynamics. This result holds across a panel of the 100 largest U.S. metropolitan areas and survives placebo tests designed to rule out mechanical correlation.

Table 1: Rent Outcomes by Density Level and Changes

Initial Rental Density	Density Change Regime	Mean Rent Change (bps)	Observations
High-Density	Crowding	6	588
High-Density	Expanding	17	412
Low-Density	Crowding	0	462
Low-Density	Expanding	-5	538

Notes: Markets are classified by initial rental household density (above / below-median) and by subsequent changes in renter household density (crowding or expanding). Mean rent change reflects average next-period real rent growth in excess of that year's average. It is reported in basis points. "Crowding" denotes $\Delta\text{RDI} \geq 0$ (density rising or unchanged); "Expanding" denotes $\Delta\text{RDI} < 0$ (density falling). The pattern is consistent with a Wheaton–DiPasquale stock-flow adjustment framework: in high-density (tight) markets, de-densification reflects the release of pent-up household formation into scarce space and is associated with stronger subsequent rent growth (17 bps) relative to continued crowding (6 bps). In low-density (slack) markets, both regimes produce near-zero or negative rent outcomes (approximately 0 and -5 bps respectively), consistent with slack absorbing adjustment pressure before it is reflected in prices.

Third, we illustrate the practical relevance of this framework through a set of empirical applications that distinguish temporary adjustment from persistent demand pressure, with implications for investors, developers, and policymakers assessing rental market conditions.

The remainder of the paper proceeds as follows. Section 2 reviews related literature on housing demand measurement. Section 3 describes the data and construction of the density measures. Section 4 presents the empirical strategy and main results. Section 5 evaluates forecasting and validation exercises. Section 6 concludes by discussing implications and limitations.

2 Background and Literature Review

A substantial literature in urban and housing economics has sought to measure rental demand using structural modeling, hedonic estimation, and affordability thresholds. Price elasticity estimates, often based on income and housing cost shares, provide insight into how demand responds to price movements under equilibrium assumptions Green, Malpezzi, and Mayo (2002); Malpezzi and Green (1996). Hedonic models explain rent levels as a function of unit characteristics and neighborhood amenities, while housing affordability metrics attempt to measure stress via cost-to-income ratios. Though valuable, these approaches often require detailed microdata that are unavailable across time and space or rely on equilibrium assumptions that abstract from short-run adjustment frictions, limiting their usefulness in settings with constrained supply or delayed price response.

A related strand of research emphasizes stock-flow dynamics in housing markets, in which quantities adjust slowly while demand conditions evolve more rapidly Goodman (1992); Wheaton (1991). In this framework, prices need not clear the market instantaneously; instead, adjustment may occur through non-price margins when supply is slow-moving or adjustment costs are high. This perspective is central to understanding rental markets during periods of rapid population growth, migration, or credit-induced shifts in tenure choice, when observed rents may lag underlying demand pressure.

Within applied real estate analysis, the multifamily housing sector has traditionally relied on indicators such as occupancy rates and net absorption to characterize demand. These measures are intuitive and operationally convenient but are tightly linked to the existing housing stock. Occupancy, by definition, cannot exceed 100 percent, and absorption is mechanically bounded by the number of new units delivered Gabriel and Nothaft (2001); Mueller and Laposa (1999). As a result, such metrics are poorly equipped to detect latent demand—periods in which prospective renters are unable to form independent households or must cohabitate due to limited supply or rising costs Pyhrr et al. (1999); Sirmans et al. (1991). This limitation has become more acute in recent years as algorithmic revenue management systems increasingly target stabilized occupancy thresholds, further muting variation and diminishing their informational content Calder-Wang and Kim (2024).

We recognize our supposition may be a gap between academic and empirical viewpoints. Perhaps large homebuilders rely on other measures of demand, and the limitations of occupancy and absorption do not serve to obfuscate their view of demand. To evaluate this, we interviewed attendees of the 2026 National Multifamily Housing Conference. While these interviews are inherently anecdotal, they are illustrative. We spoke with Adrian Foley, who is the Chief Executive Officer of Brookfield Residential. This is the homebuilding group within Brookfield, one of the world's largest alternative asset managers, with over USD 1 Trillion under management. Mr. Foley's group focuses on developing single-family homes for two purposes: selling (Build-to-Sell, or BTS) and renting (Build-to-Rent, or BTR). We posed to him the question, "How do you determine the quantity of units to build in a given site or metro?" His response was "It's driven off population growth, jobs growth, and household formation. We overlay that with affordability and you get housing demand by price. It's science-driven but very artsy in execution." We view this response in two ways. First, it suggests that homebuilders do not consider occupancy and absorption to the extent we supposed. However, in their place, they opt for slower-moving more demographic-based metrics. Is it because more housing-based measures do not contain sufficient information, or because the other measures–population, jobs, and household–growth contain even more information than occupancy and absorption? A full evaluation of the information contained in those variables is beyond the scope of this investigation, but we do compare their predictive power to that of the RDI in section 5. We are affirmed that Mr. Foley did not reference absorption or occupancy explicitly, but instead referenced variables that are unbounded. This does not mean that there is no power in occupancy or absorption, but it does suggest that experts could benefit from a more housing-specific variable that indicates demand without the mechanical constraints of other variables.

Prior research has long recognized that rental market adjustment occurs in part through household formation and crowding behavior. Studies such as Gabriel and Nothaft (2001) and Sirmans et al. (1991) document how constrained rental markets influence living arrangements, vacancy duration, and rent dynamics. However, this literature typically treats household formation outcomes as implicit or secondary margins of adjustment rather than as observable indicators of accumulated market pressure. As a result, changes in household size and crowding are often interpreted as outcomes of tight markets rather than as state variables that summarize the degree of imbalance preceding price adjustment.

Parallel work in urban economics has examined density primarily in geographic terms, such as people or dwellings per square mile, and linked it to productivity, wages, and agglomeration effects Duranton and Puga (2004); Glaeser and Maré (2001). More recent studies explore the role of densification in shaping housing affordability and urban form Ahlfeldt and Pietrostefani (2019); Albouy and Lue (2015). While informative about land use and spatial efficiency, geographic density does not directly capture household formation or space consumption behavior and may remain stable even as living arrangements adjust within the existing housing stock.

Our work reframes density through a behavioral and dynamic lens by defining it at the rental household level. We introduce the Rental Density Index (RDI), defined as the number of renters per occupied rental unit. Rather than measuring spatial intensity, RDI captures how households adjust their consumption of housing space when demand outpaces supply. In markets where renters prefer more space, increases in RDI reflect household consolidation driven by constrained conditions, while declines indicate de-densification as pressure eases. Crucially, these adjustments can occur even when occupancy remains effectively fixed, allowing RDI to reveal latent demand pressure in fully occupied markets.

We further distinguish between the level of rental crowding and short-run changes in household density. The level of RDI, after absorbing persistent city characteristics and common annual conditions via fixed effects, captures a slow-moving state variable summarizing accumulated imbalance between population growth and housing supply. Short-run changes in RDI capture contemporaneous densification that may temporarily absorb pressure without immediately translating into rent growth. This distinction aligns naturally with a stock-flow framework in which supply responds gradually to persistent imbalances rather than to transitory shocks, and maps directly onto the levels and differenced terms of our panel ARDL specification.

RDI's contribution is therefore not simply that it is unbounded, but that it captures a fundamentally different margin of adjustment. Unlike vacancies or absorption, which are constrained by the existing stock, and unlike prices, which may adjust with a lag, household density directly reflects how renters respond to scarcity at prevailing prices. In this sense, RDI provides information about market tightness that is orthogonal to traditional stock-based indicators and complementary to price-based measures.

In sum, our contribution is to reinterpret the familiar concept of density as household density, using a renter-centered indicator that makes household formation and space consumption an explicit part of rental market analysis. By linking crowding levels and short-run densification to subsequent rent growth within a panel ARDL framework, we bridge reduced-form forecasting with behavioral interpretation, with practical relevance for investors, planners, and policymakers.

3 Data and Descriptive Statistics

The data for our analysis come from three primary sources: the U.S. Census Bureau, Costar, and the U.S. Bureau of Labor Statistics. A summary of the key variables used in our analysis is provided in Table 2.

The data used to calculate RDI are from the Public Use Microdata Statistics (PUMS). The PUMS data is an offering of the U.S. Census in conjunction with the American Community Survey. The data allow for visibility into myriad aspects of a respondent's life (including the number of people in their household and whether they rent or own their home), anonymized for privacy and standardized to create a comprehensive view of the entire country. This data is tabulated by the IPUMS USA Ruggles et al. (2025). The Rental Density Index (RDI), is calculated by summing the number of people living in rental units within a CBSA, and dividing that by the sum of occupied rental units within a CBSA. Excluded from both the numerator and denominator are: group quarters (institutions); vacant units (for sale; for rent; rented or sold but not-yet-occupied); units for seasonal, recreational or other occasional use; and units for migratory workers. We compute its year-over-year difference—ΔRDI—to reflect short-term demand dynamics. This measure avoids the bounded structure of traditional demand indicators like occupancy and absorption, which are upper bounded by 100%.

Our market rent data and inventory growth data is sourced from Costar. This includes nominal effective rent per square foot and inventory units. Our inflation measures come from the U.S. Bureau of Labor Statistics (Series CPIAUCSL). We restrict our analysis to the 100 largest U.S. metropolitan statistical areas (CBSAs) by multifamily inventory as of 2005, ensuring robust sample representation and consistency over time.

To adjust for inflation, we convert nominal rent figures into real terms by subtracting the annual Consumer Price Index change from the annual rent growth change. Although CBSA-level inflation indices excluding rent would be ideal, such data are unavailable at a consistent and granular level. We carefully considered the feasibility of constructing CBSA-level inflation adjustments. We explored both FHFA metropolitan house price indexes and HUD Fair Market Rent series as potential candidates. FHFA indexes measure transaction prices in the owner-occupied housing market and are dominated by single-family sales, making them conceptually mismatched to rental rent growth in multifamily markets. HUD Fair Market Rents, while rental-based, are policy benchmarks designed for voucher administration; they incorporate smoothing, lagged information, and national CPI inputs. They are not intended to measure year-to-year market rent inflation. Because neither source provides a consistent, market-based measure of local rental inflation over our sample period, we instead remove common inflationary effects by using the national CPI.

The Costar data references the same geographic regions as our IPUMS data, as defined by the 2013 Core Based Statistical Areas. Geographic equivalence is an important consideration when joining datasets from different sources, especially when working in a spatio-temporal context like ours. We are assured of equivalency based on the adherence of IPUMS and Costar to the Census-defined TIGERweb Geography files. These files are provided in compliance with the 1998 Section 508 amendment of the Workforce Rehabilitation Act of 1973, § 1194.22 Web-based Intranet and Internet Information and Applications.

To standardize our rent-growth figures, we subtract each year's median rent growth value from each CBSA's value in that year. This gives our rent growth an annual mean near zero. This dampens effects caused by the whipsaw in pricing seen during and after the COVID-19 crisis. These normalized rent growth figures are used in Section 5, while the real (not normalized figures) are provided to the ARDL model, which uses time-based fixed effects to control for inflation and other common shocks.

Table 2: Key variables used in the empirical analysis

Variable	**Source**	**Description**
`rental_households`	U.S. Census	Number of renter-occupied housing units in an CBSA
`rental_household_bedrooms`	U.S. Census	Total number of bedrooms in renter-occupied housing units in an CBSA
`rental_population`	U.S. Census	Number of people living in rental dwellings within an CBSA
`CPI`	BLS	Consumer Price Index for All Urban Consumers: All Items, U.S. City Average
`effective_rent_growth_12_m`	CoStar	Effective rent growth in a market over the prior 12 months
`inventory`	CoStar	Total number of rental units in an CBSA
`bedroom_density_rented`	U.S. Census	Total number of renters in an CBSA divided by the total number of bedrooms in rented households
$\%\Delta$`CPI`	Calculation	December-to-December percent change in `CPI`
`RDI`	Calculation	`rental_population` divided by `rental_households`
Δ`RDI`	Calculation	Year-over-year change in `RDI`
`real_rent_growth`	Calculation	`effective_rent_growth_12_m` minus $\%\Delta$`CPI`
`relative_real_rent_growth`	Calculation	`real_rent_growth` less that year's median `real_rent_growth`
`supply_growth`	Calculation	Year-over-year percentage change in `inventory`

3.1 Rental Density Index Construction

The PUMS data is large, and complex, though very well documented. The output of the raw data requires parsing to aggregate to the CBSA level. This section describes that aggregation process.

3.1.1 Filtering and Labeling

Records are filtered to include only non-vacant units and non-group quarters GQ. This ensures that the analysis captures conventional household formation behavior rather than institutional or non-household living arrangements.

$$\text{VACANCY} = 0 \tag{1}$$

$$\text{GQ} \notin \{3, 4, 5, 6\} \tag{2}$$

Tenure is translated from numeric codes to descriptive labels as follows:

$$\text{OWNERSHP} = \begin{cases} \texttt{"OWNED"} & \text{if code} = 1 \\ \texttt{"RENTED"} & \text{if code} = 2 \\ \texttt{"OTHER"} & \text{otherwise} \end{cases} \tag{3}$$

3.1.2 Grouping Records

Next, the data are aggregated in two stages. At the household level, we sum person weights (PERWT) to obtain household population counts, retain the household weight (HHWT) as the number of households, and compute total bedrooms (BEDROOMS) by weighting unit bedroom counts by household weights.

For each household h, year y, metro area m, and tenure t we compute:

$$\text{POP}_{h,y,m,t} = \sum_{i \in h} \text{PERWT}_i \tag{4}$$

$$\text{HH}_{h,y,m,t} = \text{HHWT}_h \tag{5}$$

$$\text{BEDROOMS}_{h,y,m,t} = \text{BEDROOMS}_h \times \text{HHWT}_h \tag{6}$$

where PERWT_i is the person weight for individual i, and HHWT_h is the household weight taken from the first individual in the group.

These household-level quantities are then aggregated to the metropolitan-year-tenure level, yielding total population, total households, and total bedrooms by ownership status within each metropolitan area and year.

$$\text{POP}_{y,m,t} = \sum_{h \in (y,m,t)} \text{POP}_{h,y,m,t} \tag{7}$$

$$\text{HH}_{y,m,t} = \sum_{h \in (y,m,t)} \text{HH}_{h,y,m,t} \tag{8}$$

Table 3: Summary statistics

	N	mean	std	min	Median	max
year	2,000	2,014	5.77	2,005	2,014	2,024
inventory_units	2,000	127,489	189,877	14,696	54,792	1,457,169
supply_growth	1,900	0.02	0.02	-0.02	0.01	0.16
rent_growth	2,000	0.02	0.03	-0.16	0.02	0.32
real_relative_rent_growth_next_year	2,000	0.00	0.02	-0.12	0.00	0.28
population_owned	2,000	1,311,841	1,592,831	115,542	729,340	11,420,211
population_rented	2,000	704,440	1,127,270	76,565	323,210	9,041,819
households_owned	2,000	473,933	549,249	47,432	273,765	4,063,756
households_rented	2,000	286,193	438,032	38,095	136,014	3,744,508
density_owned	2,000	2.70	0.20	2.18	2.66	3.38
density_rented (RDI)	2,000	2.40	0.28	1.77	2.35	3.40
density_rented_change	1,900	-0.01	0.09	-0.40	-0.00	0.41
bedroom_density_rented	2,000	0.78	0.08	0.60	0.77	1.09
bedroom_density_owned	2,000	0.64	0.04	0.54	0.63	0.82

$$\text{BEDROOMS}_{y,m,t} = \sum_{h\in(y,m,t)} \text{BEDROOMS}_{h,y,m,t} \tag{9}$$

3.1.3 Feature Construction

Densities by year, metro, and tenure are calculated at the household and bedroom level.

$$\text{DENSITY}_{y,m,t} = \frac{\text{POP}_{y,m,t}}{\text{HH}_{y,m,t}} \tag{10}$$

$$\text{BEDROOM\_DENSITY}_{y,m,\text{rent}} = \frac{\text{POP}_{y,m,\text{rent}}}{\text{BEDROOMS}_{y,m,\text{rent}}} \tag{11}$$

$$\text{RDI}_{y,m} \equiv \text{DENSITY}_{y,m,t=2} \tag{12}$$

$$\Delta\text{RDI}_{y,m} = \text{RDI}_{y,m} - \text{RDI}_{y-1,m} \tag{13}$$

3.2 Summary Statistics

Table 3 provides descriptive statistics for the key variables across the full panel. On average, the `density_rented` (RDI) across all CBSAs and years is 2.4 persons per rental unit, with a standard deviation of 0.3. The average annual growth in rental supply is 2.0%, while the average `rent_growth` is 2.5%, when we remove the effects of inflation and normalize the value to the median rent growth of the year, the `real_relative_rent_growth_next_year` drops to a mean of zero. The `bedroom_density` (people-per-bedroom) in rental households is 0.12 higher than in the owned-home counterpart. The missing records exist only in the beginning or ending years as the prior period and next-period values do not exist. The latest date for which Census data exists is 2024, while the latest full-year of rent data is from 2025.

3.3 Coverage and Representativeness

The sample includes 2,000 CBSA-year observations, covering a mix of large coastal cities, fast-growing Sun Belt metros, and slower-growing Midwestern regions. Together, these CBSAs account for over 32 million rental dwellings, housing over 73 million residents, providing a true representative snapshot of national rental dynamics. In our Empirical Strategy and Results, we evaluate the degrees to which our model specification varies when applied to sparsely populated areas, outside of our coverage.

3.4 Preliminary Observations

Across the 100 CBSAs in the study, the RDI ranges from 1.77 (2020: Albany-Schenectady-Troy, NY) to 3.40 (2012: Riverside-San Bernardino-Ontario, CA). This wide range reflects both housing market constraints and regional variation in family formation and living preferences. But the RDI level alone cannot distinguish whether units are full by necessity (housing shortage) or by choice, so we capture fixed effects of time and location to account for lifecycle preferences. We also introduce the density-per-bedroom variable from the Public Use Microdata in order to evaluate these effects. In the next section we investigate how these measures interact.

4 Empirical Strategy and Results

4.1 Panel ARDL Specification

We estimate a single-stage panel ARDL in which next-year real rent growth is regressed on lagged levels and contemporaneous changes of household density and supply growth, together with lagged rent growth as an autoregressive term. For metropolitan area m in year t, the specification is:

$$\begin{aligned}
\Delta R_{m,t+1} = \alpha_m &+ \gamma_t \\
&+ \lambda \cdot \mathrm{RDI}_{m,t} \\
&+ \delta \cdot \Delta\mathrm{RDI}_{m,t} \\
&+ \phi \cdot S_{m,t} \\
&+ \psi \cdot \Delta S_{m,t} \\
&+ \rho \cdot \Delta R_{m,t} \\
&+ \kappa \cdot \mathrm{BD}_{m,t} \\
&+ \varepsilon_{m,t},
\end{aligned} \tag{14}$$

where α_m denotes metropolitan fixed effects and γ_t denotes year fixed effects. $\Delta R_{m,t+1}$ is real rent growth in metropolitan area m over year $t+1$. $\mathrm{RDI}_{m,t}$ is the Rental Density Index (renters per occupied rental unit) measured at the end of year t. $S_{m,t}$ is the supply growth rate in year t. $\mathrm{BD}_{m,t}$ is the average number of renters per rented bedroom, included to control for persistent and time-varying differences in household composition across metropolitan areas. Markets differ structurally in average household size, reflecting family formation patterns, age composition, and cultural preferences rather than market pressure. And $\mathrm{BD}_{m,t}$ ensures that these compositional differences do not confound the demand-pressure signal in $\mathrm{RDI}_{m,t}$. Δ denotes the within-entity first difference. $\varepsilon_{m,t}$ is an idiosyncratic error term, and standard errors are clustered at the metropolitan level.

The specification follows the ARDL representation of an error-correction model. The lagged level $\mathrm{RDI}_{m,t}$ plays a role analogous to an error-correction term in a traditional ECM, capturing accumulated demand pressure, though we note that because the series are stationary rather than integrated, this is properly understood as a stationary ARDL rather than a cointegrating relationship. Conditional on short-run dynamics, λ captures the extent to which above-average crowding predicts subsequent rent adjustment. The lagged level of supply growth $S_{m,t}$ enters analogously, capturing the dampening effect of accumulated new supply on rent growth. The differenced terms $\Delta\mathrm{RDI}_{m,t}$ and $\Delta S_{m,t}$ capture contemporaneous short-run changes in each variable, while $\Delta R_{m,t}$ controls for rent growth momentum. κ captures the influence of household composition on the RDI signal: metropolitan areas with structurally larger households, where more occupants share each bedroom by family preference rather than by market pressure, would otherwise appear more crowded in the RDI measure than market conditions alone would imply. By conditioning on $\mathrm{BD}_{m,t}$, the specification ensures that λ reflects genuine demand-pressure variation in household density rather than persistent compositional differences across cities. Metropolitan fixed effects absorb time-invariant differences in structural crowding levels across cities; year fixed effects absorb common aggregate shocks including national interest rate cycles and macroeconomic conditions.

Prior to estimation, we confirm that neither $\mathrm{RDI}_{m,t}$ nor $\Delta R_{m,t+1}$ exhibits unit root behaviour. Using an Im-Pesaran-Shin-style panel unit root test, which averages individual ADF t-statistics across the 100 metropolitan areas, we observe t-stats of -3.07 and -3.25 respectively (versus critical values of -2.98 for both), consistent with stationarity. This satisfies spurious regression concerns and confirms that the ARDL levels terms are not acting as cointegrating vectors, but rather as stationary predictors of a stationary outcome. Single-stage estimation is therefore appropriate and avoids the generated-regressor problems that would arise from a two-stage approach. This also confirms that the lagged level of RDI enters as a stationary predictor rather than a cointegrating vector, so the specification is an ARDL rather than an error-correction model in the Engle-Granger sense.

4.2 Results

Table 4 reports the results. The lagged level of the Rental Density Index enters with a positive and statistically significant coefficient ($\lambda = 0.0463, p < 0.01$), consistent with the stock-flow interpretation: markets where renter household density is elevated relative to the national average, after absorbing time-invariant metropolitan characteristics and common annual conditions, experience stronger real rent growth in the subsequent year. An one standard deviation increase in RDI is associated with a 130 basis point-increase in real rent growth in the next year, conditional on supply, rent dynamics, and household choices at the bedroom level.

The contemporaneous change in RDI, ΔRDI, enters with a positive and statistically significant coefficient ($\delta = 0.0343, p < 0.01$), indicating that short-run densification carries meaningful predictive content for rent growth beyond the accumulated level. This result refines the stock-flow interpretation: while the level of crowding captures persistent demand pressure, the flow of new household formation into rental markets represents an independent and contemporaneous signal of tightening conditions. Holding all else equal, a one standard deviation increase in ΔRDI is associated with a 30 basis point-increase in next-year real rent growth.

On the supply side, both the lagged level and contemporaneous change in supply growth enter with negative and statistically significant coefficients ($\phi = -0.1988, p < 0.01$; $\psi = -0.0891, p < 0.05$). The larger magnitude of the level term relative to the change term mirrors the RDI result: accumulated supply exerts a stronger dampening effect on rent growth than does contemporaneous construction activity alone. Markets that have experienced sustained inventory growth face meaningfully lower subsequent rent appreciation, even after controlling for short-run supply changes. The bedroom density control enters with a negative and statistically significant coefficient ($\kappa = -0.0846, p < 0.05$). This is consistent with the compositional role of $\mathrm{BD}_{m,t}$ described in Section 4.1: metropolitan areas where renters structurally occupy more occupants per bedroom, reflecting family formation patterns and household preferences rather than market pressure, tend to have lower subsequent rent growth once aggregate demand intensity is held fixed. As confirmed by the partial regression of $\mathrm{BD}_{m,t}$ on next-year rent growth after removing metropolitan and year fixed effects, the coefficient is driven by cross-sectional differences in household composition across cities rather than within-metro variation over time. The opposing signs on RDI and BD are therefore not contradictory but complementary: RDI captures dynamic demand pressure within markets over time, while BD absorbs persistent compositional differences across markets that would otherwise conflate family-structure variation with the latent-demand signal.

The within-R^2 of 0.141 is substantial relative to comparable one-period-ahead rent growth specifications in the literature, where values below 0.10 are typical for panel models with two-way fixed effects. The inclusion of lagged rent growth as an autoregressive term accounts for a meaningful share of this explanatory power, reflecting well-documented momentum in short-run rental price dynamics. The Ljung-Box statistic fails to reject the null of no residual autocorrelation at lags 1-4. With T=18 time periods per entity, the test has limited power at higher lags. But lags 1–4 show no evidence of misspecification.

4.3 Stock-Flow Interpretation

This section provides a simple stock-flow motivation for the empirical specification used in the paper. The objective is not to develop a fully structural model of rental markets, but rather to clarify why renter household density naturally enters a rent-adjustment equation as a lagged-level predictor — playing a role analogous to an error-correction term — in a setting with slow-moving supply

Table 4: Panel ARDL: Predictors of Next-Year Real Rent Growth

	(1) Real rent growth (next year)
Rental Density Index	
$\text{RDI}_{m,t}$ (lagged level)	0.0463***
	(0.0122)
$\Delta\text{RDI}_{m,t}$ (change)	0.0343***
	(0.0123)
Supply growth	
$S_{m,t}$ (lagged level)	−0.1988***
	(0.0420)
$\Delta S_{m,t}$ (change)	−0.0891**
	(0.0449)
Rent dynamics	
$\Delta R_{m,t}$ (lagged rent growth)	0.2832***
	(0.0259)
Affordability constraints	
$\text{BD}_{m,t}$ (bedroom density)	−0.0846**
	(0.0427)
Constant	−0.0425
	(0.0280)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Observations	1,800
Entities	100
Time periods	18
R^2_{within}	0.1405
F-statistic (robust)	38.423
p-value	0.000
Ljung-Box p-value (lag 1)	0.458

Notes: The dependent variable is real rent growth in metropolitan area m over year $t+1$. $\text{RDI}_{m,t}$ is the Rental Density Index (renters per occupied rental unit). $S_{m,t}$ is the supply growth rate. $\text{BD}_{m,t}$ is the average number of renters per rented bedroom, capturing affordability constraints at the intensive margin of rental demand. Lagged levels enter as error-correction terms within the ARDL framework; differenced terms capture short-run dynamics. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. Ljung-Box statistic tests for residual autocorrelation at lag 1 (null: no autocorrelation). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Let $D_{m,t}$ denote renter household density in metropolitan area m at time t, defined as renters per occupied rental unit. Households have a preferred or long-run density level $D^*_{m,t}$, which reflects demographic composition and structural housing characteristics that vary across metropolitan areas. When housing supply adjusts slowly, actual density may deviate from this benchmark, with the deviation representing accumulated demand pressure that has not yet been resolved through prices.

In a stock-flow framework, rents respond gradually to accumulated imbalance rather than instantaneously clearing the market. A simple partial-adjustment relationship for real rent growth can be written as:

$$\Delta R_{m,t+1} = \lambda D_{m,t} + \delta \Delta D_{m,t} + \rho \Delta R_{m,t} - \phi S_{m,t} - \psi \Delta S_{m,t} + \varepsilon_{m,t+1}, \qquad (15)$$

where $\lambda > 0$ captures the speed at which elevated household density is translated into subsequent rent adjustment, $\delta > 0$ captures the independent contribution of contemporaneous densification, and the supply terms capture the dampening effect of new inventory on that adjustment. Metropolitan fixed effects absorb persistent cross-city differences in structural density levels, so that identification relies on within-metro variation in RDI over time — precisely the variation that reflects departures from each market's long-run baseline.

This formulation highlights two complementary features of the RDI terms. The level term $D_{m,t}$ functions as a state variable: it summarizes cumulative demand pressure that has not yet been fully reflected in prices, and its predictive content is inherently forward-looking — elevated household density predicts future rent growth because it reflects imbalances that are resolved only gradually as supply and prices adjust. The differenced term $\Delta D_{m,t}$ captures a distinct and contemporaneous signal: short-run household formation into rental markets that tightens conditions independently of the accumulated level. The significance of both terms in estimation is consistent with a stock-flow process in which both the stock of latent demand pressure and its current rate of change carry information about subsequent price adjustment.

Importantly, this framework does not imply that household density causes rent growth in a structural sense. Rather, it formalizes why RDI contains information about future rent adjustment in markets where supply responds slowly and households adjust along non-price margins.

4.4 Robustness Checks

4.4.1 Placebo Tests: Nominal Rent Growth

To ensure our model is not simply picking up spurious correlations between RDI and rent growth, we conduct a placebo test using nominal rent growth as the dependent variable. If the relationship between RDI and real rent growth is driven by underlying demand pressure rather than mechanical correlation, we would expect the explanatory power of the model to decline when using nominal rent growth. Note that the coefficients should all remain unchanged, as subtracting the same value (inflation) by year, impacts all terms in the same way. To test this, we re-estimate the ARDL specification with nominal rent growth as the outcome and find that the R^2_{within} drops from 0.141 to 0.116; and the R^2_{Overall} drops from 0.103 to 0.0869. This decline in explanatory power confirms that the relationship between RDI and rent growth is not simply a spurious correlation, but rather reflects genuine predictive content about real rent dynamics.

4.4.2 Placebo Tests: Randomized RDI

To evaluate the degree to which the baseline results reflect a spurious correlation rather than an economically meaningful relationship between household density and

Placebo test results: distribution of randomized RDI coefficients

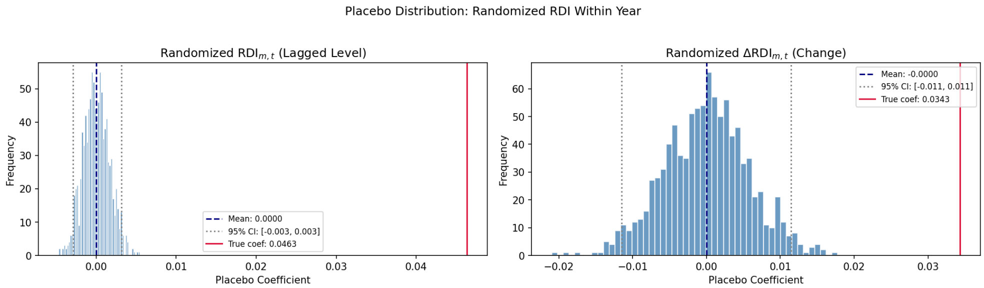

Fig. 2: Distribution of coefficients on randomized $\mathrm{RDI}_{m,t}$ (left) and $\Delta\mathrm{RDI}_{m,t}$ (right) from 1,000 placebo iterations. The vertical dashed line represents the mean of the placebo distribution, dotted lines indicate the 95% empirical interval, and the solid red line shows the true baseline coefficient.

subsequent rent adjustment, we implement a randomization placebo test. In each of 1,000 iterations, we randomly reassign $\mathrm{RDI}_{m,t}$ and $\Delta\mathrm{RDI}_{m,t}$ across metropolitan areas within each year, preserving the cross-sectional distribution of each variable in every period while severing its link to each metropolitan area's rent outcome. We then re-estimate the panel ARDL specification on the randomized data and record the coefficients on both $\mathrm{RDI}_{m,t}$ and $\Delta\mathrm{RDI}_{m,t}$.

The results are reported in Figure 2. In both panels, the placebo distributions are tightly centered at zero, with empirical means of approximately 0.000 in each case. For the lagged level, the 95% empirical interval is $[-0.003, 0.003]$; the true coefficient of 0.0463 lies more than fifteen times the upper bound of this interval and is not approached by any of the 1,000 randomized estimates. For the contemporaneous change, the 95% empirical interval is $[-0.011, 0.011]$; the true coefficient of 0.0343 exceeds the upper bound by a factor of three and similarly falls outside the entire placebo distribution.

These results confirm that neither finding is an artifact of spurious within-year correlation between RDI and rent growth. The predictive content of both the level and flow of household density reflects genuine cross-metropolitan variation in demand pressure — accumulated and contemporaneous — rather than a mechanical relationship driven by the cross-sectional structure of the data.

4.4.3 Specification Diagnostics: Stationarity and Serial Correlation

The validity of single-stage ARDL estimation rests on two conditions: that the levels terms entering the specification are stationary, and that the dynamic structure of the model is sufficient to eliminate residual serial correlation. We verify both.

As reported in Section 4, an Im-Pesaran-Shin-style panel unit root test yields average ADF t-statistics of -3.07 for $\mathrm{RDI}_{m,t}$ and -3.25 for $\Delta R_{m,t+1}$, both below the individual-series 5% critical value of -2.98. This confirms that both series are stationary, ruling out spurious regression concerns and confirming that the lagged level terms enter as stationary predictors rather than cointegrating vectors. Single-stage estimation is therefore appropriate; a two-stage approach would introduce generated-regressor bias without the super-consistency justification that cointegration would provide.

To verify that the ARDL lag structure adequately captures the dynamic adjustment process, we test for residual serial correlation using the Ljung-Box statistic at lag lengths 1 through 4. In all cases we fail to reject the null of no autocorrelation (lag 1: $p = 0.458$; lag 2: $p = 0.676$; lag 3: $p = 0.785$; lag 4: $p = 0.111$), providing no evidence of dynamic misspecification in the estimated model.

4.4.4 Dynamic Ordering Tests

To assess whether the relationship between household density and rent growth reflects a genuine forward-looking signal rather than a mechanical or reverse relationship, we estimate two dynamic ordering tests. The first replicates the baseline ARDL specification and confirms that RDI predicts future rent growth in the forward direction. The second reverses the dependent and key independent variables, asking whether current rent growth predicts future RDI. If the relationship is genuinely directional — running from accumulated demand pressure to subsequent price adjustment — we would expect the forward specification to be significant and the reverse specification to be insignificant.

The results, reported in Table 5 and Table 4, confirm this pattern. In the forward direction, both the lagged level and contemporaneous change in RDI enter with positive and statistically significant coefficients ($\lambda = 0.0463$, $p < 0.01$; $\delta = 0.0343$, $p < 0.01$), consistent with the baseline results. In the reverse direction, the coefficient on current rent growth is economically negligible and statistically indistinguishable from zero (-0.0225, $p = 0.82$), providing no evidence that rent growth predicts subsequent household density. Taken together, these results confirm that the predictive relationship runs from renter household density to rent growth, and not the reverse.

Two ancillary results from the reverse specification are consistent with the stock-flow framework. First, the lagged RDI level enters with a positive and significant coefficient (0.138, $p < 0.01$), confirming that household density is a slow-moving and persistent state variable — precisely the property that motivates its role as an error-correction term in the forward specification. Second, bedroom density enters positively and significantly (0.732, $p < 0.01$), reflecting the structural co-movement between household-level and bedroom-level crowding across metropolitan areas.

5 Value Added by the Rental Density Index in Forecasting Rent Growth

This section evaluates whether the Rental Density Index (RDI) provides incremental predictive information about future rent growth beyond commonly used indicators of rental market conditions, including vacancies, absorption, supply growth, and lagged rent growth. In contrast to the regression results presented earlier, which establish in-sample relationships consistent with a stock–flow adjustment framework, the analysis here focuses explicitly on forecast performance. All exercises are designed

Table 5: Dynamic Ordering Test 2: Rent Growth Predicting Future RDI

	(1) $\text{RDI}_{m,t+1}$ (next-year RDI)
Key variable of interest	
$\Delta R_{m,t}$ (current rent growth)	-0.0225
	(0.0999)
Controls	
$\text{RDI}_{m,t}$ (lagged level)	0.1383***
	(0.0311)
$S_{m,t}$ (lagged level)	0.0401
	(0.1734)
$\Delta S_{m,t}$ (change)	0.1866
	(0.1213)
$\text{BD}_{m,t}$ (bedroom density)	0.7317***
	(0.1009)
Constant	1.4936***
	(0.1029)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Observations	1,700
Entities	100
R^2_{within}	0.3168
F-statistic (robust)	17.973
p-value	0.000

Notes: Reverse dynamic ordering test. The dependent variable is next-year RDI in metropolitan area m. The key variable of interest is current real rent growth; its insignificance ($p = 0.82$) confirms that rent growth does not predict future household density, ruling out reverse causation as an explanation for the baseline results. The significant coefficient on lagged RDI reflects the persistence of household density as a slow-moving state variable. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

to be information-set consistent and to mirror how market participants would form expectations in real time.

We present three complementary exhibits, all out-of-sample. The first evaluates short-term, one-period-ahead predictive power in a direct horse race against standard indicators. The second examines whether persistent changes in RDI contain information about medium-term rent performance. The third provides an internal consistency check, relating the sign and magnitude of RDI changes to subsequent rent outcomes using only variation within the RDI measure itself.

5.1 Short-Term Horse Race: One-Period-Ahead Forecasts

We begin by assessing the short-term predictive power of RDI relative to alternative demand indicators. Using rolling OLS regressions, we generate one-period-ahead forecasts of real relative rent growth for each metropolitan area. In each rolling window, the forecasting model is estimated using only data available up to that point in time. We then evaluate each model's ability to sort markets into future outperformers and underperformers. In this section we refer to "outperform/er/ance" as having excess real relative rent growth. Conversely "underperform/er/ance" refers to a market or markets with lower-than-average real rent growth.

Rather than emphasizing coefficient magnitudes, we focus on ranking performance, which is both robust to scale differences across predictors and directly relevant for investors and policymakers. Specifically, for each forecast year we rank metropolitan areas by predicted rent growth and compare realized outcomes between the top and bottom quintiles, as well as between the top and bottom deciles.

The results are reported in Figure 3. Forecasts based on RDI combined with lagged rent growth consistently outperform those based on any single alternative indicator, including vacancies, absorption, supply growth, or lagged rents alone. Moreover, the RDI-based forecast also outperforms a specification that includes all four alternative indicators jointly. This pattern holds across both quintile and decile comparisons, indicating that the result is not driven by extreme observations.

These findings suggest that RDI captures a dimension of latent demand pressure that is not subsumed by traditional stock-based or flow-based measures, even when those measures are used together. In the short run, household crowding appears to summarize information about market tightness that is otherwise only weakly reflected in vacancies, absorption, or recent price movements.

5.2 Medium-Term Predictive Power and Persistence

While the previous exercise focuses on short-run rent adjustment, RDI is conceptually a slow-moving state variable reflecting accumulated imbalances between household formation and housing supply. We therefore examine whether persistent changes in RDI contain information about medium-term rent performance.

To do so, we compute, for each metropolitan area, the trailing five-year average of the sign of annual RDI changes. For example, if Los Angeles experienced four years of negative RDI changes and one year of positive RDI change, then its value would be 0.6 (the average of -1,-1,-1,-1,1). This measure captures whether a market has been experiencing sustained densification or de-densification, abstracting from short-term volatility. Metropolitan areas are then sorted into two groups: those with above-median and below-median values. We examine those groups mean real relative rent growth over the next five years.

Figure 4 reports the results of this exercise and compares the performance of the RDI-based grouping to analogous groupings constructed using the same strategy applied to vacancies, absorption, supply growth, and lagged rent growth. Using the identical grouping rule across variables, the RDI-based measure produces a clearer and more monotonic separation between future outperformers and underperformers than any of the alternatives.

This evidence indicates that sustained changes in renter household density contain information about medium-term rent dynamics that is not captured by more transitory indicators. Persistent crowding appears to reflect cumulative demand pressure that is

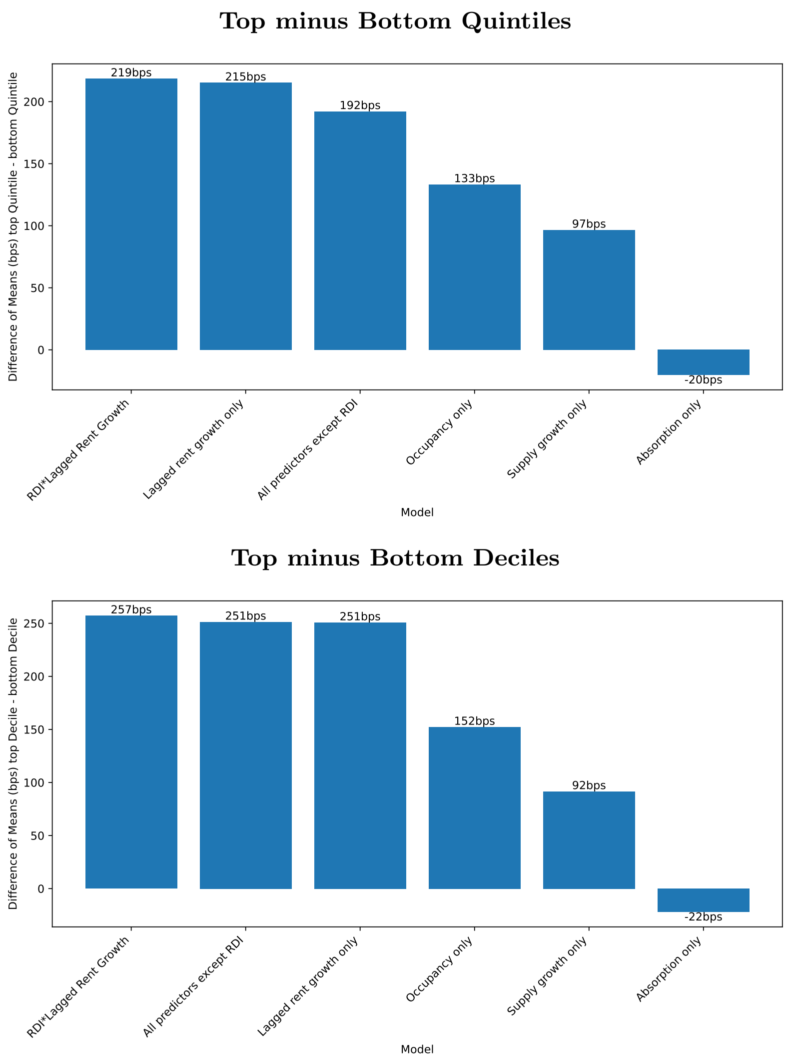

Fig. 3: Short-Term Forecast Horse Race. Difference of means: next-period real relative rent growth in forecasted high-growth and low-growth markets. Markets are ranked each period using rolling OLS forecasts based on individual predictors and selected combinations. Bars report average differences between the top and bottom quintiles (top panel) and top and bottom deciles (bottom panel). The RDI-based forecast (RDI plus lagged rent growth) outperforms all single-variable forecasts and a specification including vacancies, absorption, supply growth, and lagged rent growth jointly.

released only gradually into prices, consistent with a stock-flow adjustment process in which supply responds slowly to demand shocks.

Medium-Term Rent Performance by Persistent RDI Changes

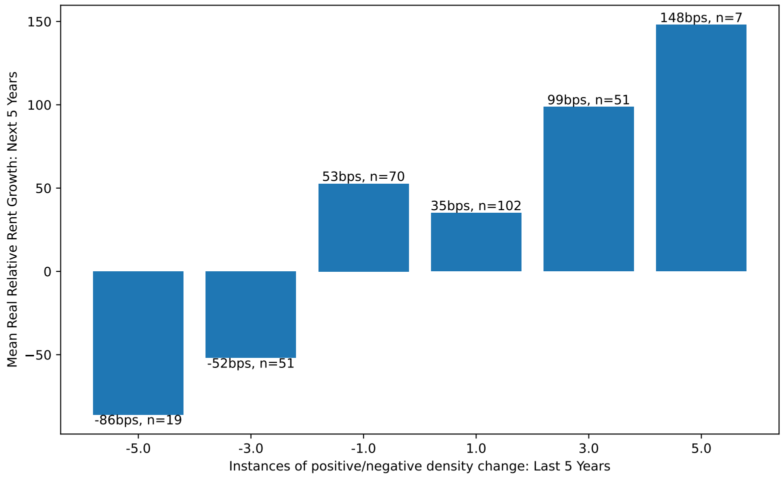

Fig. 4: Average real relative rent growth over the subsequent five years for metropolitan areas grouped by their trailing five-year sum of RDI-change sign. For example, a market that experienced five years of negative RDI growth would be represented in the leftmost bin. That market's next five years of real relative rent growth is summed then averaged with its cohort (other markets who saw five consecutive periods of negative RDI growth). The study is performed out-of-sample, with RDI readings taken in 2005-2009 (2010-2014, 2015-2020) and rent reported for 2010-2014 (2015-2019, 2020-2024)

5.3 Internal Consistency: RDI Changes and Future Rent Growth

Finally, we examine the internal consistency of the RDI signal by relating changes in RDI to subsequent rent outcomes using only variation within the RDI measure itself. This exercise is not intended as a horse race against other predictors, but rather as a credibility check on whether RDI behaves in a manner consistent with its interpretation as a measure of latent demand pressure.

Specifically, we present a histogram where bins are determined based on the sum of the sign of RDI changes over a trailing five-year window. We then examine average rent growth over the subsequent five years. The results, shown in Figure 5, reveal a clear relationship: markets with more persistently positive RDI changes subsequently experience higher average rent growth, while markets with sustained declines in RDI experience weaker rent performance.

Importantly, this relationship is evaluated out of sample, with grouping based solely on past RDI changes and outcomes measured in future periods. The near-monotonic pattern reinforces the interpretation of RDI as a meaningful summary of underlying demand pressure rather than a noisy or purely contemporaneous statistic.

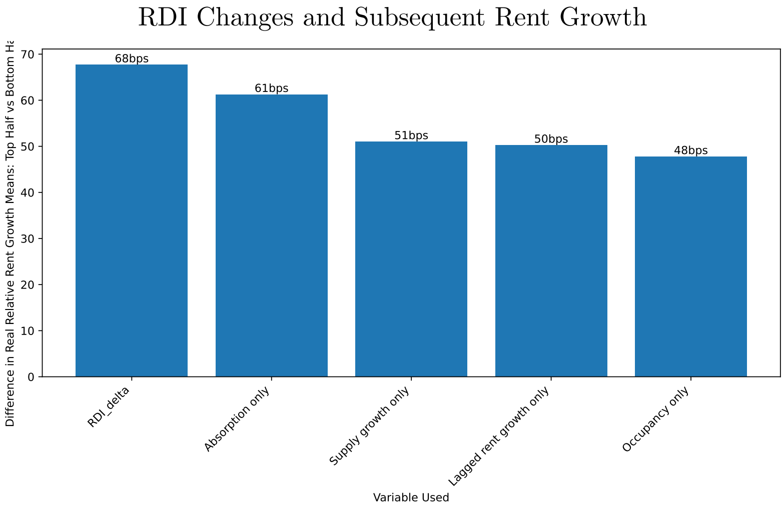

Fig. 5: Average real relative rent growth over the subsequent five years for metropolitan areas grouped by their trailing five-year value of various metrics. We split markets into two groups based on their past RDI, average occupancy, average rent growth, etc. We then take the average real relative rent growth of those groups and plot the differences. The study is performed out-of-sample, with readings taken in 2005-2009 (2010-2014, 2015-2020) and rent reported for 2010-2014 (2015-2019, 2020-2024)

5.4 Summary

Taken together, these three exercises demonstrate that RDI provides incremental predictive content beyond standard indicators of rental market conditions. RDI improves short-term forecast performance in direct horse races, captures medium-term dynamics associated with persistent market imbalance, and exhibits internally consistent relationships with future rent outcomes. These findings support the interpretation of renter household density as a behavioral margin through which latent demand pressure accumulates and is ultimately reflected in rental prices. While renter household density is informative for forecasting and market characterization, it should not be interpreted as a structural measure of demand or welfare without additional identifying assumptions.

6 Conclusion

This paper introduces the Rental Density Index (RDI), a household-level measure of rental market tightness that captures how renters adjust their consumption of housing space when supply responds slowly to changes in demand. By reframing density as the number of occupants per renter household rather than as a geographic concept, we highlight household consolidation as a key non-price margin through which rental markets absorb pressure prior to price adjustment.

Using panel data for the 100 largest U.S. metropolitan areas, we show that the level of renter household density — after conditioning on bedroom density, supply growth, and fixed effects for metropolitan area and year — contains predictive information about subsequent real rent growth. Interpreted within a stock-flow adjustment framework, the RDI level functions as a slow-moving state variable that captures accumulated imbalance between household formation and housing supply. Markets where renter household density is elevated relative to the national average experience stronger subsequent rent adjustment as prices and quantities gradually respond.

Several limitations of this approach warrant explicit discussion. First, RDI is most informative in markets where space consumption is a binding margin of adjustment. In lower-density markets, where households can adjust more readily through vacancies, migration, or physical expansion, household density contains little incremental information beyond supply growth. Second, RDI is a medium-run indicator by construction. Its predictive content weakens during periods of extreme disruption—such as the Global Financial Crisis or the COVID-19 pandemic— when institutional shocks, policy interventions, and abrupt behavioral changes dominate normal adjustment dynamics. Finally, RDI abstracts from within-market heterogeneity: it summarizes average household density and does not capture distributional differences across income groups, neighborhoods, or housing types.

The results also clarify the scope of causal interpretation. The level of the Renter Density Index should not be interpreted as a causal determinant of rent growth, nor as a structural measure of housing demand or welfare. Rather, it is a reduced-form diagnostic that summarizes how markets absorb demand pressure when supply adjusts slowly. Moving from prediction to causal inference would require exogenous variation that shifts household formation or crowding independently of expected rent dynamics—for example, plausibly exogenous migration shocks, policy-induced changes in household eligibility or living arrangements, or instruments that affect household consolidation without directly affecting rental supply or pricing. Such identification strategies lie beyond the scope of this paper but represent a promising direction for future research.

Despite these limitations, the findings have substantive implications for policy and market analysis. For policymakers, rising renter household density may signal accumulating pressure even in markets where rents appear temporarily stable or vacancies remain low. Monitoring household density alongside prices and vacancies can help distinguish whether affordability pressures are being absorbed through consolidation rather than resolved through new supply. This distinction is particularly relevant for evaluating the short-run effects of zoning reforms, construction pipelines, or demand-side subsidies, which may influence household formation behavior before they affect observed rents.

For investors and market participants, measures of renter household density provide a complementary indicator of market tightness that can improve assessment of future rent risk, particularly in supply-constrained and stabilized markets where traditional demand metrics are muted. More broadly, the results underscore that how households share space matters for understanding rental market dynamics.

Overall, this paper shows that household density is not merely a descriptive statistic but a meaningful component of housing market adjustment. By incorporating renter household density into a stock-flow view of rental markets, we provide a new lens for measuring latent demand pressure and anticipating rent movements in constrained environments. Future work that combines household-level data with exogenous shocks or policy variation may further clarify the causal mechanisms underlying these adjustment processes and extend the framework to additional settings.

Acknowledgments

Declarations

Data and Materials Availability Data that comes from public sources is available upon request. Code is also available on request. Rent-growth figures come from Costar, and thus require their permission for redistribution.
Authors' Contributions All authors contributed equally
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Ethical Approval No humans or animals are involved
Consent to Participate No humans or animals are involved
Consent to Publish No humans or animals are involved
Competing Interests There are no competing interest

Appendix A Shortage Estimates

Table A1: Selected Estimates of the U.S. Housing Unit Shortage

Year	Source	Estimated Housing Shortage (Millions of Units)
2022	Corinth and Dante (2022)	20.1
2024	National Low Income Housing Coalition (2024)	7.1
2020	National Association of Realtors (2020)	5.5
2024	Brookings Institution (2024)	4.9
2025	Zillow (2025)	4.7
2022	Zillow (2022)	4.5
2021	Up for Growth (2021)	3.9
2025	Goldman Sachs (2025), price-to-income approach	3.9
2020	Freddie Mac (2020)	3.8
2024	Freddie Mac (2024)	3.7
2025	Goldman Sachs (2025), vacancy-based approach	3.1
2023	Jones (2023)	2.5
2025	Zandi et al. (2025)	2.0
2022	National Association of Home Builders (2022)	1.5

Notes: Estimates are drawn from a range of academic, industry, and policy sources and reflect differing methodologies, definitions of equilibrium, and time horizons. Across the studies listed, the mean estimated housing shortfall is approximately 5.1 million units and the median estimate is approximately 3.9 million units.

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